

Why not build AI data centers near oceans and use sea water for cooling ?

Posted by u/Parking_Bid9954 • 0 points • 15 comments

Maybe they can even be used as desalination plants.

Comments

u/Drwynyllo • 2 points • 2025-12-09 11:42 UTC

"**SIN01 is the world's first 100% renewable AI data centre using ocean water cooling** for a 1.2GW campus"

"Furthermore, the research findings have already validated the environmental approach, with India \ [Oliveira, Sustainability Manager at Start Campus\] adding: "These researchers predict that the one degree Celsius and the plume of influence is very small.

"The discharge is very concentrated in this area on the discharge area and the bay, but it diffuses a lot. So we don't release any chemicals. We are releasing warmer water, which will have some impact – but there will be some wildlife who will benefit and others that will probably migrate to other places – yet the predicted impact is minimal."

See [<https://aimagazine.com/articles/ocean-cooled-ai-the-data-centre-redefining-renewable-energy>] (<https://aimagazine.com/articles/ocean-cooled-ai-the-data-centre-redefining-renewable-energy>)

u/Chemical-Charity8333 • 1 points • 2025-12-09 10:48 UTC

Sea water will corrode those data centers in mere days

u/TechGirl_9 • 5 points • 2025-12-09 10:36 UTC

A lot of folks have this idea, and it sounds logical at first. Huge computers get hot, ocean water is cold, boom, free cooling. But saltwater is basically the villain in this story. It destroys pipes, pumps, heat exchangers, you name it. The corrosion is brutal and the maintenance costs get ridiculous fast.

Even if you solve that, you still need stable land, cheap power, crazy networking capacity, and protection from storms and flooding. Oceans are great until a hurricane decides your data center looks like a snack.

And turning the waste heat into desalinated water isn't as simple as it sounds. The heat from servers isn't nearly strong enough to boil seawater, and building extra infrastructure just to make it work costs more than it saves.

So the idea isn't stupid at all. It's just one of those things that seems perfect until the engineering realities show up and ruin the fantasy.

u/Justarah • 1 points • 2025-12-09 10:45 UTC

Would there be any negatives to building them in polar conditions then? Obviously sheltered from wind, debris and the like.

u/deceze • 2 points • 2025-12-09 10:49 UTC

Hard to find people willing to work there. Barely any or no existing infrastructure. Hard to get materials to. Hard to find existing power plants nearby. Nowhere near an existing fiber optic cable.

u/Justarah • 1 points • 2025-12-09 10:51 UTC

So what about the likes of Greenland or Nordic countries?

u/TechGirl_9 • 1 points • 2025-12-09 12:28 UTC

Hard to find labor

u/deceze • 1 points • 2025-12-09 10:58 UTC

Most of what I listed above goes for most of Greenland. Not sure Greenlanders would be happy to have a massive data center built anywhere near their more populated areas. The most populous city is Nuuk with 20k people; it's tiny, and would be dwarfed by a modern data center.

The Nordic countries might fare a bit better here, but the areas that are populated enough to build such a center are also in a climate zone there isn't all that much cooler. The much colder hinterlands are, again, way more remote, less populated, and have all the aforelisted issues.

u/Wodentinot • 1 points • 2025-12-09 10:21 UTC

It's cheaper and easier to put them where poor people live. What are they going to do about it?

u/gssupplies • 2 points • 2025-12-09 10:15 UTC

sea water is super corrosive and would probably wreck all the equipment over time.. the maintenance costs would be insane.

u/Jolly-Credit-9998 • -8 points • 2025-12-09 10:13 UTC

Seawater is terrible for cooling due to its extreme corrosiveness and operational complexity.

The real, long-term incentive is accessing the ultimate power source: the Sun. Since the Sun holds over 99.8% of the solar system's mass and energy, putting AI centers in orbit is a literal move toward maximizing power access. This shift conceptually aligns with advancing a civilization up the Kardashev Scale, closer to utilizing all available energy from its star (Type II).

u/Dapper-Lab-9285 • 6 points • 2025-12-09 10:32 UTC

I always though it was hard to radiate off heat in space. It took the James Webb telescope months to cool down using passive radiative cooling. A data center in space would require a

massive passive cooling array to go with a massive solar panel, we are fast running out of space in space.

They have put some data centers in the sea to use it to cool them.

u/ExcessivePlumbing • 6 points • 2025-12-09 10:26 UTC

Heat exchange in (near) vacuum is extremely slow.

If you could heat up your radiator up to 400K (which you can't, but let's be generous), you'd need ~ 0.7 square meters per kilowatt.

A data center can easily consume a megawatt. You will need a 700,000 square meters radiator.

Typical Musk project.

u/Petwins • 18 points • 2025-12-09 10:08 UTC

Salt water is massively corrosive, its rarely used in cooling systems because it destroys them really really fast.

u/wenevergetfar • 1 points • 2026-03-02 02:44 UTC

Desalinate it then...i still dont see the issue

u/Petwins • 1 points • 2026-03-02 04:14 UTC

Separately massively expensive and energy intensive

u/wenevergetfar • 1 points • 2026-03-02 13:39 UTC

Wont someone think of the billionaires ❤️

u/Ok-Pepper3061 • 1 points • 2025-12-09 17:31 UTC

This is why they use closed loop systems with heat exchangers instead - you can still dump the heat into the ocean without the salt water actually touching your expensive equipment